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Digital product passport - Data carriers

Passeport numérique des produits - Supports de données

Digitaler Produktpass - Datenträger

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European foreword

This document (EN 18220:2026) has been prepared by Technical Committee CEN/CLC/JTC 24 “Digital product passport – Framework and systems”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by November 2026, and conflicting national standards shall be withdrawn at the latest by November 2026.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document has been prepared under a standardization request addressed to CEN by the European Commission. The Standing Committee of the EFTA States subsequently approves these requests for its Member States.

For relationship with EU Legislation, see informative Annex ZA, which is an integral part of this document.

Any feedback and questions on this document should be directed to the users’ national standards body. A complete listing of these bodies can be found on the CEN website.

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Introduction

0.1 Background

In 2019 the European Commission set out the European Green Deal as the EU's strategic framework for sustainable growth. A central legislative instrument in that framework is the Ecodesign for Sustainable Products Regulation (ESPR), which aims to support the Union's transition to a circular, resource-efficient economy and long-term climate neutrality.

The ESPR promotes production and consumption patterns that reduce environmental impact and improve product performance across the lifecycle. To support these objectives, the ESPR sets out ecodesign requirements that improve product durability, reliability, repairability, upgradability, reusability and recyclability and that seek to reduce waste, minimize harmful substances and improve resource and energy efficiency.

Digital product passports (DPPs) are identified in the ESPR as a key enabling mechanism to make product information traceable and accessible across value chains - supporting economic operators, manufacturers, distributors, repairers, recyclers and consumers to make informed decisions. The implementation of digital product passports will be carried out progressively. Sector-specific legislation will determine the precise DPP content and requirements for individual product groups.

This staged approach enables regulatory requirements to be refined according to sectoral characteristics while maintaining coherence with the EU's environmental policy objectives and stakeholder considerations. Where applicable, DPP requirements will be consistent with other EU legislation, such as the Construction Products Regulation and the Batteries Regulation.

To support the implementation of DPPs, 8 standards have been developed:

— EN 18219 – Digital product passport – Unique identifiers

— EN 18220 – Digital product passport – Data carriers (this document)

— EN 18222 [1] – Digital Product Passport – Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability

— EN 18216 [2] – Digital Product Passport – Data exchange protocols

— EN 18221 [3] – Digital product passport – data storage, archiving, and data persistence

— EN 18223 [4] – Digital Product Passport – System interoperability

— EN 18239[1] [5] – Digital Product Passport – access rights management, information system security, and business confidentiality

— EN 18246[2] [6] – Digital Product Passport – Data authentication, reliability and integrity

0.2 Overview

This document addresses suitable data carriers and shows how they meet the requirements. ESPR [7] defines 'data carrier' as a linear barcode symbol, a two-dimensional symbol or other automatic identification data capture medium that can be read by a device. It further stipulates that delegated acts could specify one or more data carriers to be used. This document specifies common rules for how to construct automatic identification and data capture (AIDC) media to be used as data carrier to provide a link to a DPP. These rules are based on the requirements derived from the ESPR and from the subsequent

[1] Under preparation. Stage at the time of publication: prEN 18239:2025.
[2] Under preparation. Stage at the time of publication: prEN 18246:2025.

standardization request issued by the European Commission. The requirements relate to the encoding capability, the dimensional characteristics, quality, persistence and durability as applicable. This document includes data carriers and shows how they meet the requirements.

1 Scope

This document specifies requirements for data carriers used in a digital product passport system. This covers: symbology characteristics, format, error correction codes, encoding methods, printing and production quality, and durability.

This document also specifies requirements on graphical or other indicators for easy recognition of DPP data carriers and the indication on the data carrier placement, machine readability, quality checking, links between physical product and digital representation.

The following aspects are out of scope: architecture and use cases, secure elements and any other cryptographic security features.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 14443-1:2018, *Cards and security devices for personal identification — Contactless proximity objects — Part 1: Physical characteristics*

ISO/IEC 14443-2:2020, *Cards and security devices for personal identification — Contactless proximity objects — Part 2: Radio frequency power and signal interface*

ISO/IEC 14443-3:2018, *Cards and security devices for personal identification — Contactless proximity objects — Part 3: Initialization and anticollision*

ISO/IEC 14443-4:2018, *Cards and security devices for personal identification — Contactless proximity objects — Part 4: Transmission protocol*

ISO/IEC 15415:2024, *Automatic identification and data capture techniques — Bar code symbol print quality test specification — Two-dimensional symbols*

ISO/IEC 15426-2:2023, *Information technology — Automatic identification and data capture techniques — Bar code verifier conformance specification — Part 2: Two-dimensional symbols*

ISO/IEC 15693-1:2018, *Cards and security devices for personal identification — Contactless vicinity objects — Part 1: Physical characteristics*

ISO/IEC 15693-2:2019, *Cards and security devices for personal identification — Contactless vicinity objects — Part 2: Air interface and initialization*

ISO/IEC 15693-3:2019, *Cards and security devices for personal identification — Contactless vicinity objects — Part 3: Anticollision and transmission protocol*

ISO/IEC 15961-2:2019, *Information technology — Data protocol for radio frequency identification (RFID) for item management — Part 2: Registration of RFID data constructs*

ISO/IEC 15962:2022, *Information technology — Radio frequency identification (RFID) for item management — Data protocol: data encoding rules and logical memory functions*

ISO/IEC 16022:2024, *Information technology — Automatic identification and data capture techniques — Data Matrix bar code symbology specification*

ISO/IEC 18000-3:2010, *Information technology — Radio frequency identification for item management — Part 3: Parameters for air interface communications at 13,56 MHz*

ISO/IEC 18000-63:2021, *Information technology — Radio frequency identification for item management — Part 63: Parameters for air interface communications at 860 MHz to 960 MHz Type C*

ISO/IEC 18004:2024, *Information technology — Automatic identification and data capture techniques — QR code bar code symbology specification*

ISO/IEC 18046-1:2011, *Information technology — Radio frequency identification device performance test methods — Part 1: Test methods for system performance*

ISO/IEC 18046-2:2020, *Information technology — Radio frequency identification device performance test methods — Part 2: Test methods for interrogator performance*

ISO/IEC 18046-3:2020, *Information technology — Radio frequency identification device performance test methods — Part 3: Test methods for tag performance*

ISO/IEC 18047-63:2023, *Information technology — Radio frequency identification device conformance test methods — Part 63: Test methods for air interface communications at 860 MHz to 960 MHz*

ISO/IEC 18092:2023, *Telecommunications and information exchange between systems — Near Field Communication Interface and Protocol 1 (NFCIP-1)*

ISO/IEC 19762:2025, *Information technology — Automatic identification and data capture (AIDC) techniques — Vocabulary*

ISO/IEC 21481:2021, *Information technology — Telecommunications and information exchange between systems — Near field communication interface and protocol 2 (NFCIP-2)*

ISO/IEC 29158, *Automatic identification and data capture techniques — Bar code symbol quality test specification — Direct part mark (DPM)*

EN 18219, *Digital product passport — Unique identifiers*

EPC Tag Data Standard V2. 3, https://ref.gs1.org/standards/tds/

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 19762:2025 and the following apply.

— ISO Online browsing platform: available at http://www.iso.org/obp

— IEC Electropedia: available at http://www.electropedia.org/

3.1
additional software
additional application, program, or tool that a user will install or access separately from the default setup of most smartphones or similar devices to interact with the digital product passport

EXAMPLE Proprietary applications, plugins or extensions, dedicated apps, specialized middleware, or customized software readers.

Note 1 to entry: Standard functionalities, such as web browsers, camera-based QR readers, and universal communication protocols (e.g. DNS, HTTP, HTTPS), are not considered as additional software.

3.2
barcode
printed data carrier indicating either a reference to a linear or a 2D matrix

Note 1 to entry: It excludes RFID data carriers.

3.3
batch
subset of a *model* (3.10) that is grouped by an economic operator based on common characteristics

3.4
consumer
individual member of the general public purchasing or using goods, property or services for private purposes

[SOURCE: EN ISO 14025:2010, 3.16]

3.5
data carrier
device or medium used to store data as a relay mechanism in an automatic identification and data capture system

3.6
digital product passport
DPP
digital record of product characteristics throughout its life cycle that is accessible via electronic means through a data carrier

Note 1 to entry: Example characteristics include environmental sustainability, environmental impact, and recyclability

3.7
economic operator
manufacturer, authorized representative, importer, distributor, dealer or fulfilment service provider that places the product onto the market

3.8
item
single unit of a *model* (3.10)

3.9
life cycle
consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal

[SOURCE: EN ISO 14040:2006, 3.1]

3.10
model
version of a product of which all units share the same technical characteristics and the same model identifier

3.11
NFC
near field communication
subset of HF passive radio-frequency identification with low range

3.12
persistence
process to keep a unique identifier associated with an identified object throughout its life cycle

3.13
PIA
RFID Privacy Impact Assessment Process

3.14
product
any physical good that is placed on the market or put into service

3.15
radio-frequency identification
RFID
identification technology that uses electromagnetic fields to automatically identify and track tags attached to objects

3.16
RAIN RFID (UHF RFID) per ISO/IEC 18000-63
passive UHF *radio-frequency identification* (3.15)

3.17
two-dimensional symbol
code representing data in machine-readable form by a collection of polygonal or circular cells in a regular pattern which are read optically by scanning

3.18
unique identifier
string of characters that is required to be unique among all identifiers used for all objects for a specific purpose

Note 1 to entry: A unique identifier refers to unique product identifier, unique economic operator identifier and unique facility identifier.

[SOURCE: ISO 29404:2015, 3.26, modified — "identifier" replaced by "string of characters", Note 1 to entry removed and a new Note 1 to entry added]

3.19
X dimension
specified width of the narrow in a bar code symbol or the specified width of a single element in a two-dimensional symbol

[SOURCE: ISO/IEC 19762:2025, 3.2.1.10]

4 Concepts

4.1 DPP objective

The general objective of the DPP is to ensure that all value chain stakeholders have access to essential, traceable, and reliable product information. This is done through a link to the web or to a lookup mechanism associated with the product. The identifier is represented in a data carrier that can be read automatically. In cases where the lookup mechanism through an identifier is temporarily not available, and there is sufficient space (barcode) or memory (RFID tag), additional information may be made available in the data carrier.

4.2 Basic specifications

This document specifies the basic rules applicable to data carriers for the identification of model, batch, and item level products. Additional legislation determines precisely the level of identification (model, batch, item) and the data carriers that apply to a given product group.

4.3 General considerations

4.3.1 New and non-new products

The DPP will apply to different product groups available on the market. The circular economy will facilitate products to be sold, used, re-sold and re-used multiple times. In some cases, the identifier and associated data carrier will remain unchanged through the lifecycle of the product. In other cases, if a new digital product passport is needed, then a new identifier and the associated data carrier will be required. A data carrier should remain usable after several used-reused-repair cycles having an impact on the durability of the support of the data carrier associated to the product.

4.3.2 DPP and other applications

Unique identification and associated data carrier technologies are widely used in some sectors for trade and logistic applications, consumer communication via the web and support for other regulatory requirements. Depending on the requirements set for a given product group, it may be that the existing data carriers related to the product meet the DPP requirements, at least partially. The data carrier specification for DPP should thus take into consideration the state of identification and data carriers' usage in the product group and this usage includes production specifications established by existing industry standards (e.g. barcode module size, print quality and location, RFID tag frequency or memory constraints). This is to say that DPP implementation will not necessarily imply the use of a new data carrier.

4.3.3 Data carrier users and reading devices

It is expected that the DPP concept will be used by many parties, where consumers are a key target. Consumers will have the ability to discover relevant information about a product, which can influence the purchasing decision. After purchase, the DPP will enable finding useful information such as user manuals, safety instructions or guidelines for recycling.

The DPP will also be used by manufacturers, distributors, resellers, recyclers, etc., for example to verify the availability and accuracy of information related to products, and to enable the reuse, repair, and recycling of products. In addition, governmental agencies, such as customs, will use the DPP to access relevant information about products.

The devices used to read the data carrier and access DPP information are smart devices and other AIDC devices (e.g: industrial barcode scanners and RFID readers in the supply chain).

5 Requirements

5.1 General

This clause describes the requirements that data carriers should meet, as stipulated in the ESPR regulation. A product item shall have at least one data carrier physically present either on the product itself, on the packaging or on documentation in accordance with this document.

5.2 Data encoding

5.2.1 Data content

A data carrier shall encode a unique product identifier that allows to access the DPP. The data carrier shall comply with the rules of the unique ID to enable a smooth interoperability between systems, including the syntax when used.

5.2.2 Data syntax

Data syntaxes of unique identifier are defined in EN 18219.

For examples of data syntax, see Annex A.

5.2.3 Character set

The data carrier shall support character sets as required by the identifier schemes defined in EN 18219. Other data may be encoded using characters specified in ISO/IEC 8859-1 [10] or ISO/IEC 646 [11].

5.3 Data carrier reading

5.3.1 Reading process for barcodes

Where individual products are scanned in hand, typical scan distances would be 10 cm to 30 cm. Where automated, distances will vary based on many factors which may impact the size specification for the barcode and typically affect either large products (i.e. home appliance, furniture, machine) or product groupings like master cartons or pallets. In such cases, the reading distance will generally be up to 30 cm. The reading distance of the DPP data carrier will typically range between 2 and 30 cm. In general, for barcodes, this distance can vary depending on several factors, such as barcode quality, lighting conditions, barcode size.

5.3.2 Reading process for NFC

NFC technologies operate in a one-to-one communication mode between a reader and a tag. The DPP data carrier reading process is thus like the process used when the DPP data carrier is a bar code.

5.3.3 Reading process for RAIN RFID (UHF RFID) per ISO/IEC 18000-63

RAIN RFID (UHF RFID) per ISO/IEC 18000-63 is designed to read multiple tags simultaneously over several meters. Different techniques can be used to read only a specific tag, such as filtering by communications protocol, adjusting the power output of the reader and the placement of the antenna, filtering by software or letting the user choose (in a list).

5.3.4 Decoding software

For end consumer usage, the unique product identifier shall be useable to access DPP public information without the need to register, download DPP-specific software, or add user credentials. For smart devices, this implies that the decoding software should be natively available in the operating system.

5.4 Marking or embedding methods on product, packaging, labelling or associated document on product

5.4.1 General

To support the persistence of the data carrier, it is important to consider the marking method(s) and whether the data carrier is placed on the product, embedded in the product, on the packaging, and/or documentation accompanying the product. It is possible that the mark on the packaging, on the product or embedded in the product, or on the documentation is lost, damaged or removed, this would result in the loss of the data carrier linking the physical product to its identifier. The allowed methods are 5.4.2, 5.4.3, 5.4.4, 5.4.5 and 5.4.6.

5.4.2 Marking data carrier on product item

This method refers to the process of permanently marking or engraving information directly onto a part or product. There are several techniques that could be used to permanently identify a product throughout its lifetime.

5.4.3 Marking data carrier on packaging

The representation of the identifier with a data carrier is often part of the packaging of the product. This approach is used commonly for consumer goods.

It is important to consider a marking method that supports persistence of the identifier in the data carrier for the lifecycle of the product.

NOTE As it is possible the original packaging could become damaged or lost, this would result in the loss of the data carrier linking the physical product to its identifier.

5.4.4 Labelling

Another method is to attach a label with the data carrier on the product or on the product packaging.

The economic operator should assess the process of printing, affixing, or integrating the data carrier onto the packaging or physical item, considering factors that may affect its durability and persistency.

5.4.5 Document

The DPP product identifier may be included in a document accompanying the product. The document may be physical or digital.

If the documentation is the only source of the DPP product identifier the data carrier shall be included in the documentation accompanying the product.

— Examples of physical documentation: User manual, warranty documents, customer printed receipt, and conformity documents.

— Examples of digital documentation: E-commerce web page, Digital customer receipt, online customer account, and e-mail.

5.4.6 Embedded

A data carrier used for digital product passport can be embedded into the product itself. Examples: textile products, tyres.

5.5 Dimensional characteristics for barcodes

5.5.1 Error correction

Error correction code (ECC) is a crucial feature that helps 2D symbols, such as QR Codes and Data Matrix codes, remain readable even if they are partially damaged, obscured, or dirty. ECC involves adding extra data to the barcode, a higher error correction level will in theory make the code more robust, but it will also make the code larger. The error correction level will not necessarily make a barcode easier to read, if the barcode print quality is poor.

5.5.2 Data carrier size

Several factors influence the size of the data carrier such as:

— the type of symbology;

— the X dimension that applies for the application;

— The selected error correction rate for data carrier technologies that support error correction;

— the amount of data encoded;

— the encoding methods of characters and character sets;

— the size of the product to which the DPP data carrier applies;

— the print method and the media.

Different industries and applications may have specific standards or recommendations for the X-dimension to ensure compatibility with their scanning and printing equipment. The X-dimension shall be chosen carefully to ensure that the barcode can be scanned accurately by the intended scanning environments (e.g. Point-of-Sale, consumer mobile devices). The combination of the above factors will determine the size of the data carrier. Opting for a smaller X-dimension to reduce the overall size based on the required encoded data may lead to a higher-density barcode, which can be challenging for lower-resolution scanners to read. The X-dimension also impacts printing. Printers need to have the appropriate resolution to produce clear and distinct bars or modules at the chosen X-dimension.

5.6 Data carrier quality/performance

5.6.1 General

The DPP data carrier shall reach a minimum level of quality to ensure performance in the targeted environment(s). This subclause specifies the standards that define implementation of different data carrier technology.

5.6.2 Two-dimensional symbols

Where minimum print quality for 2D symbols is assessed, the methodology shall conform to ISO/IEC 15415:2024. This standard defines the methodology for grading the quality of printed 2D barcodes. The grade includes a grade level, measuring aperture and the wavelength of light used for the measurement.

The minimum print quality grade is typically 1,5/80 %/660, where:

Minimum print quality grades are established based upon the scanner operating environment (e.g. retail point-of-sale, transport and logistics, smart device) but these specifications, established within industry application standards follow a convention of

— 1,5 is the overall symbol quality grade;

— 80 % is the measuring aperture related to the measured X dimension;

— illumination 660 is the peak response wavelength in nanometres.

ISO/IEC 29158, Direct Part Mark (DPM) Quality Guideline shall be used to assess the symbol quality of direct marked parts, where the mark is applied directly to the surface of the item and the reading device is a two-dimensional imager.

There are many industry standards that specify, per application scanning/reading environment, data carrier specifications such as minimum and maximum size, minimum print quality, and/or symbol placement.

5.6.3 Barcode verification

In the case the quality/performance is to be assessed, the quality/performance of 2D barcode symbols shall use ISO/IEC 15426-2:2023.

5.6.4 HF, NFC and RAIN RFID (UHF RFID) per ISO/IEC 18000-63

In case the quality/performance is to be assessed, the quality/performance of RFID (UHF or NFC) shall use the following standards:

— ISO/IEC 18046-1:2011: Information technology — Radio frequency identification device performance test methods — Part 1: Test methods for system performance;

— ISO/IEC 18046-2:2020: Information technology — Radio frequency identification device performance test methods — Part 2: Test methods for interrogator performance;

— ISO/IEC 18046-3:2020: Information technology — Radio frequency identification device performance test methods — Part 3: Test methods for tag performance;

— or NFC Forum tag performance: Test Cases for Tag Performance (Version 1.1).

There may be additional standard requirements adapted to specific product types such as ISO 20909 [12], ISO 20910 [13], ISO 20911 [14], ISO 20912 [15] and ISO/PAS 25091 [16] for tyres.

Data carriers should be write protected after encoding the unique identifier as specified in EN 18219.

5.7 Data carrier design

5.7.1 Data carrier placement

To facilitate the choice of more sustainable products, the presence of DPP data carriers should be clear to the consumer. This could be achieved when the data carrier is visible (e.g. visible 2D, RFID, NFC). This could also be achieved with a mark indicating presence of RFID or NFC when they are not visible to the consumer. They should be recognizable to consumers without having to read the brand name and model number in text on the label. For placement of visible data carriers, industry standards should be followed to ensure scanning efficiency and fulfilment of existing industry requirements for safety.

5.7.2 Human readable interpretation

Human readable interpretation (HRI) is the information encoded in the DPP data carrier presented close to it in a clearly legible font, such as OCR-B specified in ISO 1073-2 [17]. Different industries may have specific recommendations for HRI. Additional legislations can specify rules applicable to different product groups.

5.7.3 Graphical symbol marking

A 2D data carrier that links to a DPP may have a graphical marking to distinguish it from other data carriers.

When an optional graphical marking is used by the economic operator, it should comply with ISO 7000 [18], IEC 60417 [19] symbol 6452 or 6452-1.

5.7.4 Accessibility

DPP data carriers shall not limit accessibility of users with a range of abilities by taking into account guidelines and standards such as Packaging - Accessible design - information and marking ISO 19809 [20] and accessibility requirements for ICT products and services EN 301549 [21] v3.2.1 for web and mobile (digital). Accessibility shall cover both digital and physical interactions with products, packaging and documentation to ensure all consumers can easily obtain DPP information.

NOTE Consumer mobile and assisted device solutions are improving accessibility by enabling users to locate products (e.g. graphical patterns adjacent to 2D barcodes) and communicate essential product features and DPP information through language localization and structured menus.

5.8 Other considerations

5.8.1 References to recognized standards

Data carrier and quality standards used for digital product passport (DPP) applications shall be recognized by ISO and IEC. Where necessary, other specifications may be considered in the following order of preference: existing European standards, national standards, and specifications developed by industry fora. The use of additional software to enhance accessibility may also be considered.

5.8.2 Proven technologies

The data carriers selected for DPP applications should have proven implementations on the market, both to produce the data carrier and for the capture and processing of the encoded data with the goal of minimizing disruption.

6 Data carrier technologies

6.1 General

This clause introduces 2D barcodes and radio-frequency technologies. It provides an overview of their technical characteristics and shows how they meet the requirements specified in Clause 5. New data carriers emerging in the future could be included in a new version of this document and be subject to the same assessment against the identification schemes requirements.

All data carriers described in Clause 6 shall be used to represent at least one of the globally unique product identifiers defined in EN 18219.

The list of requirements is expressed in the document with the normative references.

6.2 Two-dimensional barcodes

6.2.1 General

Two-dimensional (2D) barcodes store data in both the horizontal and vertical dimensions through a pattern of dots, squares, hexagons or rectangles. 2D barcodes can in theory encode up to 7 000 characters.

6.2.2 Data Matrix

Data Matrix shall conform to ISO/IEC 16022:2024. It is a compact 2D barcode that uses black and white cells arranged in a square or rectangular pattern. It can store text, numeric, or binary data and is highly efficient for encoding small amounts of information. Its robust error correction makes it ideal for marking small items like medical devices and electronics and cosmetics and also in retail since its adoption within retail scanning systems is trending rapidly.

6.2.3 QR Code

QR code (Quick Response Code) shall conform to ISO/IEC 18004:2024. It is a 2D barcode that stores text, URLs, or data. It is easily scanned by smartphones or cameras, enabling quick access to information. Known for its versatility, fast readability, and error correction, it is widely used in marketing, payments, logistics, and digital interactions. QR code adoption within retail scanning systems is trending rapidly.

6.2.4 Two-dimensional barcodes characteristics

6.2.4.1 Specific characteristics

Table 1 describes the characteristics of each 2D barcode under consideration:

Table 1 — Specific characteristics of 2D data carriers

		Data Matrix	QR Code
Character set		ISO/IEC 8859-1:1998 ISO/IEC 646:1991	ISO/IEC 8859-1:1998 ISO/IEC 646:1991
Capacity	Digits/ AN	3 116 numeric, 2 335 alphanumeric characters	7 089 numeric, 4 296 alphanumeric, 2 953 Kanji Characters
	Bits	12,44	14,81 (H) 34,36 (L)
	Bytes	up to 1 555 bytes	1 852 bytes (H) to 4 296 bytes (L)
Shall be capable of encoding URI		Yes	Yes
Shall be capable of encoding ISO/IEC 15459		Yes	Yes
Shall be capable of encoding granularity		Yes	Yes
Standard		ISO/IEC 16022:2024	ISO/IEC 18004:2024
Market adoption		Pervasive	Pervasive
Native software in smart devices		Partial	Pervasive
Error correction		Automatic. 25 % to 33 % of the symbol's data capacity	7 % to 30 % of the symbol's data capacity
Typical range of reading/scanning range		10 to 30 cm	10 to 30 cm

		Data Matrix	QR Code
Simultaneous reading/scanning		Not without additional software or specialist device	Not without additional software or specialist device
Power supply		N/A	N/A
Security		Nothing inherent when produced per the symbol specifications without specialized software or marking techniques	Nothing inherent when produced per the symbol specifications without specialized software or marking techniques
Frequency		N/A	N/A
Radio regulation		N/A	N/A

6.3 Radio frequency data carriers

6.3.1 General

There are many different RFID technologies. In this document, and for DPP-related purposes, we consider 3 of them:

— High Frequency (HF) passive RFID;

— Near Field Communication (NFC);

— RAIN RFID (UHF RFID) per ISO/IEC 18000-63.

NOTE NFC technology can be considered as a subset of HF RFID because it operates at the same frequency. Its implementation in and use with smart devices is mainly described in NFC Forum Technical Specifications.

6.3.2 HF RFID

6.3.2.1 General

HF (High Frequency) RFID technologies data carriers shall comply to: ISO/IEC 14443-1:2018, ISO/IEC 14443-2:2020, ISO/IEC 14443-3:2018, ISO/IEC 14443-4:2018, or ISO/IEC 15693-1:2018, ISO/IEC 15693-2:2019, ISO/IEC 15693-3:2019 or ISO/IEC 18000-3:2010.

6.3.2.2 General overview of HF RFID

High Frequency (HF) passive RFID is a set of RFID technologies that operate at 13,56 MHz. These technologies mainly use magnetic field coupling between readers and tags to wirelessly exchange data. They are used for proximity or vicinity applications like product identification/authentication, transport ticketing system, access control, contactless payment, and much more.

6.3.2.3 Data encoding

Data encoding of HF RFID tag shall comply to ISO/IEC 15961-2:2019, and ISO/IEC 15962:2022.

6.3.2.4 Air interface protocols

Air interface protocols for HF RFID shall comply to ISO/IEC 14443-1:2018, ISO/IEC 14443-2:2020, ISO/IEC 14443-3:2018, ISO/IEC 14443-4:2018, or ISO/IEC 15693-1:2018, ISO/IEC 15693-2:2019, ISO/IEC 15693-3:2019, or ISO/IEC 18000-3:2010.

6.3.2.5 Test methods for HF RFID air interface compliance

ISO/IEC 10373-6 [22] defines test methods which are specific to proximity cards and objects, and proximity coupling devices and proximity extended devices, defined in ISO/IEC 14443-1:2018, ISO/IEC 14443-2:2020, ISO/IEC 14443-3:2018, and ISO/IEC 14443-4:2018.

ISO/IEC 10373-7 [23] defines test methods which are specific to vicinity cards and objects, and vicinity coupling devices, defined in ISO/IEC 15693-1:2018, ISO/IEC 15693-2:2019, ISO/IEC 15693-3:2019.

ISO/IEC 18047-3 [24] defines test methods for determining the conformance of radio frequency identification devices (tags and interrogators) for item management with the specifications given in ISO/IEC 18000-3:2010.

6.3.3 NFC

6.3.3.1 General overview of NFC (HF)

NFC (Near Field Communication) is a wireless communication technology that enables devices to exchange data over short distances, typically within 4 cm. Like HF RFID, NFC operates at 13,56 MHz. NFC is designed for secure, proximity-based interactions. It supports three modes:

— reader/writer;

— card emulation; and

— peer-to-peer communication.

6.3.3.2 Air interface protocols

NFC air interface protocols shall comply to ISO/IEC 18092:2023 and ISO/IEC 21481:2021. Additional information can also be found in NFC Forum Analog and Digital Specifications [25] (type 2, type 3, type 4, type 5).

6.3.3.3 Data Encoding

Data Encoding of NFC shall comply to ISO/IEC 15961-2:2019, and ISO/IEC 15962:2022. To be natively read by smart devices, NFC Forum Technical Specifications provides ways to encode data in NFC tags [25].

6.3.3.4 Test methods for NFC air interface compliance

Compliance with NFC Forum Specifications is described in the NFC Forum Certification program [25] (NFC certification release 14).

6.3.4 RAIN RFID (UHF RFID) per ISO/IEC 18000-63

6.3.4.1 General overview of RAIN RFID (UHF RFID) per ISO/IEC 18000-63

RAIN RFID (UHF RFID) per ISO/IEC 18000-63 uses propagation radio waves to automatically identify and read multiple items simultaneously over several metres. It consists of two main components: tags, which are small devices attached to items, and readers, which emit radio waves to detect and read the tags.

This technology is widely used in industries like retail, logistics, healthcare, and manufacturing for inventory management, asset tracking, access control, and more. Unlike barcodes, RAIN RFID (UHF RFID) per ISO/IEC 18000-63 does not require line-of-sight, making it a highly efficient tool for tracking large quantities of items in real-time.

6.3.4.2 Air interface protocols

The air interface protocol for RAIN RFID (UHF RFID) per ISO/IEC 18000-63 shall comply to ISO/IEC 18000-63:2021.

NOTE GS1 Gen 2V3 [26] is fully incorporated in ISO/IEC 18000-63 and provides additional specifications for EPC/RFID.

6.3.4.3 Data Encoding

ISO/IEC 18000-63:2021 specifies the data encoding. This encoding includes the numbering system identifier toggle bit (T). If the toggle bit (T) is set to 0_2, then the application is referred to as a GS1 EPC global application and shall be as data encoding described in GS1's EPC Tag Data Standard V2.3. If the toggle bit (T) is set to 1_2, the application is referred to as an ISO unique item identifier (UII) as registered according to ISO/IEC 15961-2:2019.

6.3.4.4 Test methods for RAIN RFID (UHF RFID) per ISO/IEC 18000-63 air interface compliance

ISO/IEC 18047-63:2023 shall be used for compliance testing of tags and readers for RAIN RFID (UHF RFID) per ISO/IEC 18000-63 devices (tags and interrogators). It is also described in GS1 Certification Program [27].

6.3.5 Radio Frequency Privacy

EN 16571 [28] (RFID Privacy Impact Assessment Process) describes when to implement a PIA and how to assess the level of risk. It also describes all kind of possible threats and proposes mitigation techniques to reduce the risk to an acceptable level. It covers all kind of RFID technologies including UHF passive and NFC.

6.3.6 Radio frequency data carriers characteristics

Table 2 describes the characteristics of the radio data carrier technologies.

Table 2 — Typical use and specific characteristics of RFID data carriers

		RAIN RFID (UHF RFID) per ISO/IEC 18000-63	NFC and HF RFID
Character set		Depends on the application standard	UTF-8
Capacity	Digits/ AN	Depends on the application standard: text or binary depending on the item identifier specification and reader capability used by the application	32 to ~ 16000 UTF-8 characters
	Bits	96 bits typical ~ up to 480 bits. With optional additional user memory of several kbits	256 bits to ~ 130 kbits
	Bytes	12 to 60 bytes	32 bytes to ~16 kbytes
Shall be able of encoding URI		Yes (for large memory)	Yes
Shall be able of encoding ISO/IEC 15459		Yes	Yes

		RAIN RFID (UHF RFID) per ISO/IEC 18000-63	NFC and HF RFID
Shall be able of encoding granularity		Yes	Yes
Shall be able of encoding standard		ISO/IEC 18000-63:2021	ISO/IEC 18092:2023, ISO/IEC 21481:2021 NOTE NFC Forum also provide additional information in NFC Forum Technical Specifications [25], and ISO/IEC 14443-1:2018, ISO/IEC 14443-2:2020, ISO/IEC 14443-3:2018, ISO/IEC 14443-4:2018
Market adoption		Pervasive	Pervasive
Native software in smart devices		Partial	Pervasive
Error correction		Yes	Yes
Typical range of reading/scanning distance		Up to 10 m or more	Less than 5 cm
Simultaneous reading/scanning		Up to 1000 tags per second	No
Power supply		Tags can be battery assisted or derive power from the reader's electromagnetic field	Tags can be battery assisted or derive power from the reader's magnetic field
Security		Yes	Yes
Frequency		860–930 MHz	13,56 MHz
Radio regulation		Regulations vary by region	Widely available

Annex A
(informative)

Examples of Data Syntax

A.1 Examples of Data Syntax from ID schemes of EN 18219

ID Scheme 1: Structured path: Example of QR code encoding a GS1 identifier in an Automatic identification and data capture techniques — Encoding and resolving identifiers over HTTP compliant: GS1 Digital Link URI.

EXAMPLE 1 QR Code with the encoded data (model level): "https://example.com/01/09524000059109/22/2A/10/ABC123/21/12345XYZ?11=251121"



NOTE 1 ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.

EXAMPLE 2 QR code encoding a GS1 identifier in an ISO/IEC 18975 compliant: GS1 Digital Link URI (batch level).

Encoded data: "https://example.com/01/09524810000339/10/YA12AB"



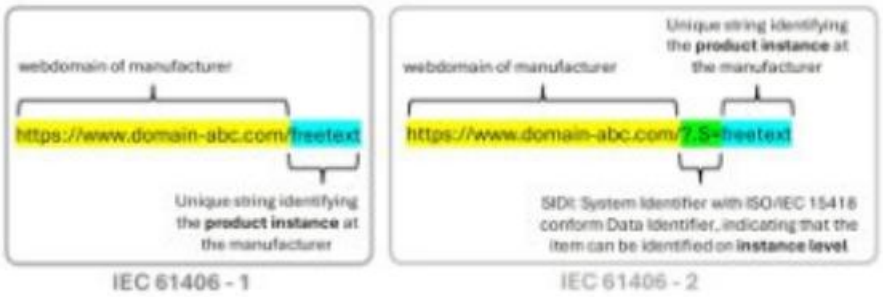
NOTE 2 ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.

EXAMPLE 3 QR code encoding a GS1 identifier in an ISO/IEC 18975 compliant: GS1 Digital Link URI (item level).

"https://www.example.com/01/09524810000339/21/12345"

NOTE 3 ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.

A.2 Examples as describes in Identification Link - Part 1: General requirements and Identification link - Part 2: Types/models, lots/batches, items and characteristics



A.3 Example of Identification Link - Part 1: General requirements and Identification link

The example of Table A.1 is leveraging Identification Link - Part 1: General requirements and Identification link as per EN IEC 61406-1 [29].

Encoded data: "https://www.example.com/freetext"

The outer graphical frame with a filled triangular corner indicates model level.

Table A.1 — Identification Link - Model



NOTE ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.

A.4 Example is leveraging Identification link - Part 2: Types/models, lots/batches, items and characteristics

The example of Table A.2 is leveraging Identification link - Part 2: Types/models, lots/batches, items and characteristics as per EN IEC 61406-2 [30].

Encoded data: "https://www.example.com?.P=freetext"

Table A.2 — Identification Link - Model, lots/batches, items



NOTE ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.

A.5 Examples of syntax with MH-10 Data Identifiers

These ideas are expanded upon to provide a fully implementable system in the AutoID specification.

Figure A.1 below for an example of the query string approach using ANSI MH10.8.2 Data Identifiers to identify a product and its expiry date

Encoded data : "https://WWW.EXAMPLE.COM?.25P = QCELMI12345&.S = 654321"



Figure A.1 — Example of the query string approach using ANSI MH-10.8.2 Data Identifiers to identify a product and its expiry date



Figure A.2

NOTE ISO/IEC 18004 [31] requires a 4X quiet zone around the QR Code.

Annex B
(informative)

Data carriers and ID schemes for products

B.1 General

With reference to digital product passport - Unique ID Schemes as specified in EN 18219, the data carriers shall be interoperable, by ensuring: *Unique Identification:* The unique product identification shall be retrievable from a data carrier as specified in this document, linking it to the digital product passport.

B.2 ID scheme 1: Web enabled, structured path and query ID for products

B.2.1 Web enabled, structured path ID for products

B.2.1.1 Web enabled, structured path ID for products in a QR Code

See Table B.1 for an example of web enabled structured path identification for products in QR Code using ISO/IEC 18975 [32] and the ISO/IEC 15459 series.

Table B.1 — ID scheme 1: Structured path QR Code

Component	Meaning	Value
Data carrier	QR-code example	
Encoded data	https://example.com/01/09524000059109/22/2A/10/ABC123/21/12345XYZ?11=251121	
NOTE ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.		

Size of web enabled structured path identification for products using a QR Code.

Table B.2 is providing examples of a QR Code depending on the chosen X-dimension for retail and point of sales scanning environment.

Table B.2 — QR Code estimation size (mm) for point of sales

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	41 × 41	16,236	20,294	40,589
M	41 × 41	16,236	20,294	40,589
Q	49 × 49	19,404	24,255	48,51
H	53 × 53	20,988	26,235	52,47

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
Module x-dimensions (mm) ➔		0,396	0,495	0,99

B.2.1.2 Web enabled, structured path ID for products in a Data Matrix

Size of web enabled structured path identification for products using a Data Matrix.

Table B.3 is providing examples of web enabled structured path identification for products in a Data Matrix using ISO/IEC 18975:2024 [32] and ISO/IEC 15459-2 [33], ISO/IEC 15459-3 [34] and ISO/IEC 15459-6 [35].

Table B.3 — ID scheme 1: Structured path Data Matrix

Component	Meaning	Value
Data carrier	Data Matrix	
Encoded data	https://example.com/01/09524000059109/22/2A/10/ABC123/21/12345XYZ?11=251121	
NOTE ISO/IEC 16022:2024 requires a 1 X quiet zone around the Data Matrix		

Table B.4 is providing the possible size of a Data Matrix depending on the chosen X-dimension and point of sales scanning environment.

Table B.4 — Size of web enabled structured path in a Data Matrix for point of sales

Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
34 × 34	13,464	16,83	33,66
Module x-dimensions (mm) ➔	0,396	0,495	0,99

B.2.2 Web enabled, query string ID for products

B.2.2.1 Web enabled, query string ID for products in a QR Code

See Table B.5 for an example of web enabled query string identification for products in a QR Code, using ASC MH10.8.2 Data Identifier (DI) per ISO/IEC 15418.

Table B.5 — ID Scheme 1: Query string QR Code

Component	Meaning	Value
Data carrier	QR Code example query string	
Encoded data	https://www.example.com/?.25P=QCELMI12345&.S=654321	
NOTE ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.		

Size of the web enabled query string identification for products in a QR code.

Table B.6 provides indications regarding different sizes of QR Codes, depending of the choice of of X-dimensions.

Table B.6 — QR Code estimation size

Error correction	Module Count including quiet zone	Minimum size (mm)	Target size (mm)	Maximum size (mm)
L	37 × 37	14,652	18,315	36,63
M	41 × 41	16,236	20,294	40,589
Q	45 × 45	17,82	22,275	44,55
H	49 × 49	19,404	24,255	48,51
Module x-dimensions (mm) ➔		0,396	0,495	0,99

B.3 ID scheme 2: Identification Link (IL) string for products

B.3.1 Identification link (IL) for products, per IEC 61406-1

Table B.7 provides an example of QR Code with Identification Link (IL) with structure and graphical frame to indicate Identification Link (IL) with structure, using IEC 61406-1

Table B.7 — Identification link (IL) without structure QR Code

Component	Meaning	Value
Data carrier	QR-code example with graphical frame to indicate item.	HERE A QR CODE WITH BLACK FILLED TRIANGLE
Encoded data	https://www.example.com/freetext	
NOTE ISO/IEC 18004:2024 requires a 4X quiet zone around the QR Code.		

Table B.8 provide indications regarding different sizes of QR Codes when using ID link, depending on the choice of X-dimensions.

Table B.8 — QR Code estimation size

QR CoError correction	Module Count including quiet zone	Including the graphical marking in Quiet Zone (additional 1X on each side)	Minimum size (mm)	Medium size (mm)	Other size (mm)
L	-	-	-	-	
M	37 × 37	39 x 39	9,75	12,95	38,61
Q	37 × 37	39 x 39	9,75	12,95	38,61
H	41 × 41	43 x 43	10,75	15,05	42,57
Module x-dimensions (mm) ➔			0,250	0,350	0,99

B.3.2 Example of QR Code with Identification Link (IL) with structure and graphical frame to indicate per IEC 61406-2

Table B.9 provides indications for Identification Link (IL) with structure, using IEC 61406-1

Table B.9 — See for an example of using IEC 61406-1, IEC 61406-2, and ASC MH10.8.2 DI per ISO/IEC 15418

Component	Meaning	Value
Data carrier	QR Code example with graphical frame to indicate item.	
Full example	https://www.example.com/?.S=freetext&.10B=freetext	
NOTE ISO/IEC 18004: 2024 requires a 4X quiet zone around the QR Code.		

Table B.10 provides indications regarding different sizes of QR Codes when using ID link with structure, depending of the choice of X-dimensions.

Table B.10

Error correction	Module Count including quiet zone	Including the graphical marking in Quiet Zone (additional 2X on each side)	Minimum size (mm)	Maximum size (mm)	Other size (mm)
L	-	-	-	-	-
M	41 × 41	45 × 45	11,25	15,75	44,55
Q	45 × 45	49 × 49	12,25	17,15	48,51
H	49 × 49	55 x 55	13,75	19,25	54,45
Module x-dimensions (mm) ➔			0,250	0,350	0,99

B.4 ID scheme 3: Decentralized identifiers (DID) for products

Table B.11 provides an example of Decentralized Identifiers (DIDs) v1.0 and did:web.

Table B.11 — Decentralized identifiers (DIDs) for products

Component	Meaning	Value
Data carrier	QR code example	
Encoded data	https://resolver.io/did:web:example.com:model4TR/?service=item-dpp	
NOTE ISO/IEC 18004: 2024 requires a 4X quiet zone around the QR Code.		

Table B.12 provides indications regarding different sizes of QR Codes, depending of the choice of of X-dimensions for a DID.

Table B.12 — QR Code estimation size (mm)

Error correction	Module Count including quiet zone	Example of small size (mm)	Example of Target size (mm)	Other example of size (mm)
L	41 × 41	16,236	20,294	40,589
M	45 × 45	17,82	22,275	44,55
Q	49 × 49	19,404	24,255	48,51
H	57 × 57	22,572	28,215	56,43
Module x-dimensions (mm) ➔		0,396	0,495	0,99

B.5 ID scheme 4: Identification for products and product groups

B.5.1 Product and group identification, RAIN RFID (UHF RFID) per ISO/IEC 18000-63

No visual representation of the encoding in an RFID tag.

See Table B.13 below for an example of identification based on ISO-registered ISO/IEC 15961-2.

Table B.13 — Identification based on ISO-registered ISO/IEC 15961-2

Component	Meaning	Value
RAIN MB01 (UII/EPC) encoding	Full encoded UII including AFI, DI, IAC, CIN, SN	AFI = 0xA1, "25SUN043325711MH8031 200000000001"
UII type (AFI = 0A1)	Indicates ISO/IEC 17360 6-bit character encoding	ISO/IEC 17360 6-bit encoded
UII purpose	Identifies a serialised product issued by an ISO/IEC 15459 registered entity	DI = 25S
Entity CIN issuing agent	ISO/IEC 15459 Issuing Agency Code (IAC)	UN (Dun and Bradstreet)
Entity CIN	Company Identification Number assigned by D&B	043325711
Serial number	Unique serial number per item	MH8031200000000001

See Table B.14 for an example of identification based on SGTIN-96 EPC (per GS1 TDS).

Table B.14

Component	Meaning	Value
GTIN	Global Trade Item Number	09506000134352
Serial Number	Unique serial number for the item	123456789
EPC Pure Identity URI	Carrier-neutral URN representation for legacy data sharing	urn:epc:id:sgtin:95060001343.05.123456789
EPC Tag URI	RFID tag-specific URN representation	urn:epc:tag:sgtin-96:3.95060001343.05.123456789
EPC Hex	Binary encoding on RFID tag, beginning with address 20h of MB01	3066C4409047E140075BCD15

B.5.2 Product and group identification, 2D-symbols

Table B.15

Component	Meaning	Value
Data carrier	QR-Code example	
Encoded data	https://example.com/?.25P=QCELMI12345&.1T=ABC123&.S=654321	
NOTE ISO/IEC 18004: 2024 requires a 4X quiet zone around the QR Code.		

See Table B.16 for size of QR Code for product and group identification

Table B.16

Error correction	Module Count including quiet zone	Example of minimum size (mm)	Example of target size (mm)	Example of maximum size (mm)
L	41 × 41	16,236	20,294	40,589
M	41 × 41	16,236	20,294	40,589
Q	45 × 45	17,82	22,275	44,55
H	49 × 49	19,404	24,255	48,51
Module x-dimensions (mm) ➔		0,396	0,495	0,99

B.6 ID scheme 5: Digital Object Identifier (DOI) for products

See Table B.17 for an example of using ISO 26324.

Table B.17 — Identification with DOI for products

Component	Meaning	Value
Data carrier	QR Code example ISO/IEC 18004: 2024 requires a 4X quiet zone around the QR Code.	
Encoded data	https://doi.org/10.1000/N44APB7E	

Component	Meaning	Value
NOTE	ISO/IEC 18004: 2024 requires a 4X quiet zone around the QR Code.	

Size for product and group identification using a QR Code.

Table B.18

Error correction	Module Count including quiet zone	Example of minimum size (mm)	Example of target size (mm)	Example of maximum size (mm)
L	33 × 33	24,519	32,67	32,67
M	37 × 37	27,491	36,63	36,63
Q	37 × 37	27,491	36,63	36,63
H	41 × 41	30,463	40,589	40,589
Module x-dimensions (mm) ➔		0,743	0,99	0,99

Annex C
(informative)

DPP data carrier recognition

C.1 Example of Data carrier multiple use implicit recognition (no need for recognition)

Table C.1 Example of QR code embedding an identifier in URI a link, 0 compliant: GS1 Digital Link URI syntax used either for consumer engagement and price look-up at point of sales, link to DPP.

Encoded data: "https://example.com/01/09524000059109/22/2A/10/ABC123/21/12345XYZ?11=251121"

Table C.1 — QR Code multiple use, implicit recognition


NOTE ISO/IEC 18004 [31] requires a 4X quiet zone around the QR Code.

C.2 Examples of data carrier with explicit recognition

An example of an explicit indication that a data carrier includes the DPP is to surround the 2D barcode by a frame and a triangularly filled corner.

This technique is specified in EN IEC 61406-1:2022 and EN IEC 61406-2:2024.

Table C.2 is an example with EN IEC 61406-1:2022.

Encoded data: "https://www.example.com/freetext"

Table C.2 — Graphical marking model

The outer graphical frame with a filled triangular corner indicates model level. 
NOTE ISO/IEC 18004 [31] requires a 4X quiet zone around the QR Code.

Table C.3 is an example from EN IEC 61406-2:2024.

Encoded data: "https://www.example.com?.P=freetext"

Table C.3 — Graphical marking batch/lot or item level

The outer graphical frame with a cut-off triangular corner indicates batch/lot or item level. 
NOTE ISO/IEC 18004 [31] requires a 4X quiet zone around the QR Code.

Annex ZA
(informative)

Relationship between this European Standard and the essential requirements of 2024/1781 aimed to be covered

This European Standard has been prepared under a Commission's standardization request M/604 AMD 1 to provide one voluntary means of conforming to essential requirements of (EU) Regulation 2024/1781.

Once this standard is cited in the Official Journal of the European Union under that (EU) Regulation 2024/1781, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding requirements of that regulation, and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Articles of (EU) Regulation 2024/1781

Essential requirements of (EU) Regulation 2024/1781 Article 9, 10 and 11	**Clause(s)/subclause(s) of this EN**	**Remarks/Notes**
Article 9, (2), (b)	5.1	
Article 9, (2), (d),	4.2, 6.1, 6.2.4	
Article 10, (1), (a)	5.2.1, 5.2.2	
Article 10, (1), (b)	5.4, 5.7.1, 5.7.2	
Article 10, (1), (c)	5.2, 5.3	
Article 10, (1), (g)	4.2	Presumption of conformity only for the access via a data carrier.
Article 12, (6), (a), (b), (c)	5.2.1	

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the products and services falling within the scope of this standard.

Bibliography

[1] EN 18222, *Digital Product Passport - Application Programming Interfaces (APIs) for the product passport lifecycle management and searchability*

[2] EN 18216, *Digital product passport - Data exchange protocols*

[3] EN 18221, *Digital product passport - data storage, archiving, and data persistence*

[4] EN 18223, *Digital Product Passport – System interoperability*

[5] EN 18239,[3] *Digital Product Passport - access rights management, information system security, and business confidentiality*

[6] EN 18246,[4] *Digital product passport - Data authentication, reliability and integrity*

[7] Ecodesign for Sustainable Products Regulation (ESPR) - https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L_202401781

[8] EN ISO 14025:2010, *Environmental labels and declarations - Type III environmental declarations - Principles and procedures (ISO 14025:2006)*

[9] EN ISO 14040:2006/A1:2020, *Environmental management - Life cycle assessment - Principles and framework - Amendment 1 (ISO 14040:2006/Amd 1:2020)*

[10] ISO/IEC 8859-1:1998, *Information technology — 8-bit single-byte coded graphic character sets — Part 1: Latin alphabet No. 1*

[11] ISO/IEC 646:1991, *Information technology — ISO 7-bit coded character set for information interchange*

[12] ISO 20909, *Radio frequency identification (RFID) tyre tags*

[13] ISO 20910, *Coding for radio frequency identification (RFID) tyre tags*

[14] ISO 20911, *Radio frequency identification (RFID) tyre tags — Tyre attachment classification*

[15] ISO 20912, *Conformance test methods for RFID enabled tyres*

[16] ISO/PAS 25091, *Angular sweep test methods for radio frequency identification (RFID) enabled tyres*

[17] ISO 1073-2, *Alphanumeric character sets for optical recognition — Part 2: Character set OCR-B — Shapes and dimensions of the printed image*

[18] ISO 7000, *Graphical symbols for use on equipment — Registered symbols*

[3] Under preparation. Stage at the time of publication: prEN 18239:2025.
[4] Under preparation. Stage at the time of publication: prEN 18246:2025.

[19] IEC 60417, *Graphical symbols for use on equipment - 12-month subscription to regularly updated online database comprising all graphical symbols published in IEC 60417*

[20] ISO 19809, *Packaging — Accessible design — Information and marking*

[21] EN 301549, *Accessibility requirements for ICT products and services*

[22] ISO/IEC 10373-6, *Cards and security devices for personal identification — Test methods — Part 6: Contactless proximity objects*

[23] ISO/IEC 10373-7, *Cards and security devices for personal identification — Test methods — Part 7: Contactless vicinity objects*

[24] ISO/IEC 18047-3, *Information technology — Radio frequency identification device conformance test methods — Part 3: Test methods for air interface communications at 13,56 MHz*

[25] https://nfc-forum.org/build/specifications

[26] GS1 Gen 2V3, https://ref.gs1.org/standards/gen2/

[27] GS1 Certification Program. https://www.gs1.org/standards/rfid/certification

[28] EN 16571, *Information technology - RFID privacy impact assessment process*

[29] EN IEC 61406-1:2022, *Identification Link - Part 1: General requirements*

[30] EN IEC 61406-2:2024, *Identification link - Part 2: Types/models, lots/batches, items and characteristics*

[31] ISO/IEC 18004:2024, *Information technology — Automatic identification and data capture techniques — QR code bar code symbology specification*

[32] ISO/IEC 18975:2024, *Information technology — Automatic identification and data capture techniques — Encoding and resolving identifiers over HTTP*

[33] ISO/IEC 15459-2, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 2: Registration procedures*

[34] ISO/IEC 15459-3, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 3: Common rules*

[35] ISO/IEC 15459-6, *Information technology — Automatic identification and data capture techniques — Unique identification — Part 6: Groupings*

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